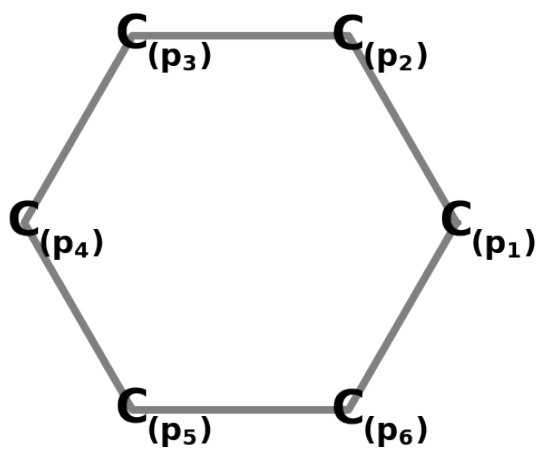


Benzene



Point Group: **D_{6h}**

p orbitals

Symmetry behavior of the p-orbitals:

$$\Gamma_{red} = 6E \oplus -2C''2 \oplus -6\sigma h \oplus 2\sigma v$$

$$\Gamma_{irred} = 1B2g \oplus 1E1g \oplus 1A2u \oplus 1E2u$$

- SALC of B2g: $\frac{\sqrt{6}(p_1 - p_2 + p_3 - p_4 + p_5 - p_6)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(2p_1 + p_2 - p_3 - 2p_4 - p_5 + p_6)}{6}$
- SALC of E1g: $\frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2}$
- SALC of A2u: $\frac{\sqrt{6}(p_1 + p_2 + p_3 + p_4 + p_5 + p_6)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(2p_1 - p_2 - p_3 + 2p_4 - p_5 - p_6)}{6}$
- SALC of E2u: $\frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{B2g} = [[\tilde{\alpha} - 2\tilde{\beta}]]$$

$$H_{E1g} = \begin{bmatrix} \tilde{\alpha} + \tilde{\beta} & 0 \\ 0 & \tilde{\alpha} + \tilde{\beta} \end{bmatrix}$$

$$H_{A2u} = [[\tilde{\alpha} + 2\tilde{\beta}]]$$

$$H_{E2u} = \begin{bmatrix} \tilde{\alpha} - \tilde{\beta} & 0 \\ 0 & \tilde{\alpha} - \tilde{\beta} \end{bmatrix}$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A2u with energy = $\tilde{\alpha} + 2\tilde{\beta}$

(

$$\begin{aligned} [1] * [1. \text{ SALC in A2u}] &= [1] * \left[\frac{\sqrt{6}p_1}{6} + \frac{\sqrt{6}p_2}{6} + \frac{\sqrt{6}p_3}{6} + \frac{\sqrt{6}p_4}{6} + \frac{\sqrt{6}p_5}{6} + \frac{\sqrt{6}p_6}{6} \right] \\ &= \frac{\sqrt{6}(p_1 + p_2 + p_3 + p_4 + p_5 + p_6)}{6} \end{aligned}$$

)

- orbital of E1g with energy = $\tilde{\alpha} + \tilde{\beta}$

(

$$\begin{aligned} \begin{bmatrix} 1 & 0 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in E1g} \\ 2. \text{ SALC in E1g} \end{bmatrix} &= \begin{bmatrix} 1 & 0 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} + \frac{\sqrt{3}p_6}{6} \\ \frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2} \end{bmatrix} \\ &= \frac{\sqrt{3}(2p_1 + p_2 - p_3 - 2p_4 - p_5 + p_6)}{6} \end{aligned}$$

)

- orbital of E1g with energy = $\tilde{\alpha} + \tilde{\beta}$

(

$$\begin{aligned} \begin{bmatrix} 0 & 1 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in E1g} \\ 2. \text{ SALC in E1g} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} + \frac{\sqrt{3}p_6}{6} \\ \frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2} \end{bmatrix} \\ &= \frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2} \end{aligned}$$

)

- orbital of E2u with energy = $\tilde{\alpha} - \tilde{\beta}$

(

$$\begin{aligned} \begin{bmatrix} 1 & 0 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in E2u} \\ 2. \text{ SALC in E2u} \end{bmatrix} &= \begin{bmatrix} 1 & 0 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} - \frac{\sqrt{3}p_6}{6} \\ \frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2} \end{bmatrix} \\ &= \frac{\sqrt{3}(2p_1 - p_2 - p_3 + 2p_4 - p_5 - p_6)}{6} \end{aligned}$$

)

- orbital of E2u with energy = $\tilde{\alpha} - \tilde{\beta}$

(

$$\begin{aligned} \begin{bmatrix} 0 & 1 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in E2u} \\ 2. \text{ SALC in E2u} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} - \frac{\sqrt{3}p_6}{6} \\ \frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2} \end{bmatrix} \\ &= \frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2} \end{aligned}$$

)

- orbital of B2g with energy = $\tilde{\alpha} - 2\tilde{\beta}$

(

$$\begin{aligned} \begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in B2g} \end{bmatrix} &= \begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{6}p_1}{6} - \frac{\sqrt{6}p_2}{6} + \frac{\sqrt{6}p_3}{6} - \frac{\sqrt{6}p_4}{6} + \frac{\sqrt{6}p_5}{6} - \frac{\sqrt{6}p_6}{6} \end{bmatrix} \\ &= \frac{\sqrt{6}(p_1 - p_2 + p_3 - p_4 + p_5 - p_6)}{6} \end{aligned}$$

)

s orbitals

Symmetry behavior of the s-orbitals:

$$\Gamma_{red} = 6E \oplus 2C''2 \oplus 6\sigma h \oplus 2\sigma v$$

$$\Gamma_{irred} = 1A1g \oplus 1E2g \oplus 1B1u \oplus 1E1u$$

- SALC of A1g: $\frac{\sqrt{6}(s_1+s_2+s_3+s_4+s_5+s_6)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(2s_1-s_2-s_3+2s_4-s_5-s_6)}{6}$
- SALC of E2g: $\frac{s_2}{2} - \frac{s_3}{2} + \frac{s_5}{2} - \frac{s_6}{2}$
- SALC of B1u: $\frac{\sqrt{6}(s_1-s_2+s_3-s_4+s_5-s_6)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{6}$
- SALC of E1u: $\frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{A1g} = [[\tilde{\alpha}_s + 2\tilde{\beta}_s]]$$

$$H_{E2g} = \begin{bmatrix} \tilde{\alpha}_s - \tilde{\beta}_s & 0 \\ 0 & \tilde{\alpha}_s - \tilde{\beta}_s \end{bmatrix}$$

$$H_{B1u} = [[\tilde{\alpha}_s - 2\tilde{\beta}_s]]$$

$$H_{E1u} = \begin{bmatrix} \tilde{\alpha}_s + \tilde{\beta}_s & 0 \\ 0 & \tilde{\alpha}_s + \tilde{\beta}_s \end{bmatrix}$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A1g with energy = $\tilde{\alpha}_s + 2\tilde{\beta}_s$

$$\begin{aligned} & (\\ & [1] * [\text{1. SALC in A1g}] = [1] * \left[\frac{\sqrt{6}s_1}{6} + \frac{\sqrt{6}s_2}{6} + \frac{\sqrt{6}s_3}{6} + \frac{\sqrt{6}s_4}{6} + \frac{\sqrt{6}s_5}{6} + \frac{\sqrt{6}s_6}{6} \right] \\ & = \frac{\sqrt{6}(s_1 + s_2 + s_3 + s_4 + s_5 + s_6)}{6} \\ &) \end{aligned}$$

- orbital of E1u with energy = $\tilde{\alpha}_s + \tilde{\beta}_s$

$$\begin{aligned} & (\\ & [1 \ 0] * \begin{bmatrix} \text{1. SALC in E1u} \\ \text{2. SALC in E1u} \end{bmatrix} = [1 \ 0] * \begin{bmatrix} \frac{\sqrt{3}s_1}{3} + \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{6} \\ \frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2} \end{bmatrix} \\ & = \frac{\sqrt{3}(2s_1 + s_2 - s_3 - 2s_4 - s_5 + s_6)}{6} \\ &) \end{aligned}$$

- orbital of E1u with energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
(

$$\begin{aligned} \begin{bmatrix} 0 & 1 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in E1u} \\ 2. \text{ SALC in E1u} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \end{bmatrix} * \left[\frac{\sqrt{3}\mathbf{s}_1}{3} + \frac{\sqrt{3}\mathbf{s}_2}{6} - \frac{\sqrt{3}\mathbf{s}_3}{6} - \frac{\sqrt{3}\mathbf{s}_4}{3} - \frac{\sqrt{3}\mathbf{s}_5}{6} + \frac{\sqrt{3}\mathbf{s}_6}{6} \right] \\ &= \frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2} \end{aligned}$$

)

- orbital of E2g with energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
(

$$\begin{aligned} \begin{bmatrix} 1 & 0 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in E2g} \\ 2. \text{ SALC in E2g} \end{bmatrix} &= \begin{bmatrix} 1 & 0 \end{bmatrix} * \left[\frac{\sqrt{3}\mathbf{s}_1}{3} - \frac{\sqrt{3}\mathbf{s}_2}{6} - \frac{\sqrt{3}\mathbf{s}_3}{6} + \frac{\sqrt{3}\mathbf{s}_4}{3} - \frac{\sqrt{3}\mathbf{s}_5}{6} - \frac{\sqrt{3}\mathbf{s}_6}{6} \right] \\ &= \frac{\sqrt{3}(2\mathbf{s}_1 - \mathbf{s}_2 - \mathbf{s}_3 + 2\mathbf{s}_4 - \mathbf{s}_5 - \mathbf{s}_6)}{6} \end{aligned}$$

)

- orbital of E2g with energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
(

$$\begin{aligned} \begin{bmatrix} 0 & 1 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in E2g} \\ 2. \text{ SALC in E2g} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \end{bmatrix} * \left[\frac{\sqrt{3}\mathbf{s}_1}{3} - \frac{\sqrt{3}\mathbf{s}_2}{6} - \frac{\sqrt{3}\mathbf{s}_3}{6} + \frac{\sqrt{3}\mathbf{s}_4}{3} - \frac{\sqrt{3}\mathbf{s}_5}{6} - \frac{\sqrt{3}\mathbf{s}_6}{6} \right] \\ &= \frac{\mathbf{s}_2}{2} - \frac{\mathbf{s}_3}{2} + \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2} \end{aligned}$$

)

- orbital of B1u with energy = $\tilde{\alpha}_s - 2\tilde{\beta}_s$
(

$$\begin{aligned} \begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} 1. \text{ SALC in B1u} \end{bmatrix} &= \begin{bmatrix} 1 \end{bmatrix} * \left[\frac{\sqrt{6}\mathbf{s}_1}{6} - \frac{\sqrt{6}\mathbf{s}_2}{6} + \frac{\sqrt{6}\mathbf{s}_3}{6} - \frac{\sqrt{6}\mathbf{s}_4}{6} + \frac{\sqrt{6}\mathbf{s}_5}{6} - \frac{\sqrt{6}\mathbf{s}_6}{6} \right] \\ &= \frac{\sqrt{6}(\mathbf{s}_1 - \mathbf{s}_2 + \mathbf{s}_3 - \mathbf{s}_4 + \mathbf{s}_5 - \mathbf{s}_6)}{6} \end{aligned}$$

)

Transitions from p orbitals into s orbitals

State with Occupation [2, 2, 2, 0, 0, 0] (0 unpaired electrons, mult 1.0)

Transition from ψ_2 into ψ_{s_2} :

- $-[2, 2, 2, 0, 0, 0], (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{3}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\frac{\sqrt{3}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 8\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 7\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-[2, 2, 2, 0, 0, 0], (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = 0, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = 0, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 8\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 7\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 1, 1, 1, 0) (4 unpaired electrons, mult 5.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 1, 1, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 1, 1, 1, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{3}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\frac{\sqrt{3}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 4\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 3\tilde{\beta} + \tilde{\beta}_s$

- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 1, 1, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 0, 1, 1, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = 0, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = 0, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 4\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 3\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 1, 1, 1, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0), (0, 0, 0, 1, 0, 0)$
- ψ_4 : linear combination = $\frac{\sqrt{3}(2p_1 - p_2 - p_3 + 2p_4 - p_5 - p_6)}{3}$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_4 : linear combination = $\frac{\sqrt{3}(2s_1 - s_2 - s_3 + 2s_4 - s_5 - s_6)}{3}$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 4\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 1, 1, 1, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0), (0, 0, 0, 0, 1, 0)$
- ψ_5 : linear combination = 0, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_5 : linear combination = 0, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 4\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 0, 1, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 0, 1, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{3}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\frac{\sqrt{3}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = 0, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = 0, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 1, 2, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0), (0, 0, 0, 0, 1, 0)$
- ψ_5 : linear combination = 0, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_5 : linear combination = 0, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 7\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 1, 0, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 1, 0, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{3}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\frac{\sqrt{3}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = 0, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = 0, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 1, 2, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0), (0, 0, 0, 1, 0, 0)$
- ψ_4 : linear combination = $\frac{\sqrt{3}(2p_1-p_2-p_3+2p_4-p_5-p_6)}{3}$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_4 : linear combination = $\frac{\sqrt{3}(2s_1-s_2-s_3+2s_4-s_5-s_6)}{3}$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 7\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 0, 1, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{3}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\frac{\sqrt{3}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 0, 1, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = 0, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = 0, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 2, 1, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0), (0, 0, 0, 0, 1, 0)$
- ψ_5 : linear combination = 0, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_5 : linear combination = 0, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 7\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 1, 0, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{3}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\frac{\sqrt{3}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 1, 0, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = 0, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = 0, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 5\tilde{\beta} + \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 2, 1, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0), (0, 0, 0, 1, 0, 0)$
- ψ_4 : linear combination = $\frac{\sqrt{3}(2p_1-p_2-p_3+2p_4-p_5-p_6)}{3}$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_4 : linear combination = $\frac{\sqrt{3}(2s_1-s_2-s_3+2s_4-s_5-s_6)}{3}$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $6\tilde{\alpha} + 6\tilde{\beta}$
- Energy of State after Excitation: $5\tilde{\alpha} + \tilde{\alpha}_s + 7\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

18. State: $A = (2, 2, 1, 1, 0, 0) \leftrightarrow B = (2, 2, 1, 1, 0, 0)$

$$E_{dispersion} = -\frac{2\tilde{\delta}^4 \left(-9(\tilde{\alpha} - \tilde{\alpha}_s) \left(\tilde{\alpha} - \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s \right) - (\tilde{\alpha} - \tilde{\alpha}_s) \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right) - 6 \left(\tilde{\alpha} - \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s \right) \right)}{r^6 (\tilde{\alpha} - \tilde{\alpha}_s) \left(\tilde{\alpha} - \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s \right) \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

Ground-State:

$A = [2, 2, 2, 0, 0, 0] \leftrightarrow B = [2, 2, 2, 0, 0, 0]$

$$E_{dispersion} = -\frac{8\tilde{\delta}^4}{r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$